

Suggested Term Project #3

Model-Free Q-Learning and Optimal Portfolio Choice (3 Assets)

Central Research Question:

Can a **model-free reinforcement learning algorithm**, which has **no knowledge of asset return moments**, learn the **theoretically optimal mean-variance portfolio allocation** purely from realized portfolio returns?

This project explicitly contrasts **economic theory** with **model-free machine learning** in a controlled financial environment.

Students will:

1. **Construct a controlled financial environment** with three assets.
2. **Estimate or calibrate** asset return moments using real data.
3. **Derive the true optimal portfolio** under mean-variance preferences.
4. **Train a model-free Q-learning agent** using only simulated returns (i.e., using the synthetic data).
5. **Compare learned portfolio policies** to the theoretical benchmark.
6. **Evaluate convergence speed and accuracy**, including under portfolio constraints.

Data-Generating Process (Treated as a Ground Truth Environment):

This is a **controlled financial environment** used to generate synthetic data for training a Q-learning algorithm (i.e., an AI trading agent) to make investment decisions.

Asset excess returns follow a stationary Gaussian process:

$$r_{t+1} = \mu + \varepsilon_{t+1}, \text{ with } \varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma)$$

- $\mu \in \mathbb{R}^3$ and $\Sigma \in \mathbb{R}^{3 \times 3}$ are **fixed and known to the students who estimate or calibrate them based on the historical daily returns of the following three traded assets in reality:**

These three assets are **three** ETFs:

- Equity: SPY (US stocks), whose daily returns can be downloaded from <https://finance.yahoo.com/quote/SPY/history/>
- Bonds: IEF (7-10y Treasuries), whose daily returns can be downloaded from <https://finance.yahoo.com/quote/IEF/history/>
- Commodity: GLD (gold), whose daily returns can be download from <https://finance.yahoo.com/quote/GLD/history/>
- Model-free Q-learning is **not allowed** to use μ or Σ in learning

Theoretical True Optimal Portfolio (Benchmark)

Assume a mean-variance investor with risk aversion γ .

If by the “standard problem” you mean the **mean–variance investor with a full-investment (budget) constraint**,

$$\max_w \mu^\top w - \frac{\gamma}{2} w^\top \Sigma w, \text{ subject to } \mathbf{1}^\top w = 1.$$

There is a closed-form solution.

Portfolio Constraints (Bonus Points)

(1) Long-only (no shorting):

$$\mathcal{W} = \{w \in \mathbb{R}^3: w_i \geq 0, \forall i\}$$

(2) Bounded weights (limited shorting / leverage):

$$\mathcal{W} = \{w \in \mathbb{R}^3: \ell_i \leq w_i \leq u_i, \forall i\}$$

Example: $\ell_i = -0.5$, and $u_i = 1.5$.

Reinforcement Learning Formulation

- **State s_t**

The state can contain **only observable information** at time t . Students can make the Q-learning stateless, that is, do not use any state variable.

- **Action a_t : Discrete Portfolio Grids**

Only need to consider w_1 and w_2 , as $w_3 = 1 - w_1 - w_2$. The discretization of the portfolio weight space is crucial.

- **Reward R_{t+1}**

Choose one primary reward, consistent with the objective specified above:

$$R_{t+1} = r_{p,t+1} - \frac{\gamma}{2} r_{p,t+1}^2$$

for portfolio p formed in period t .

- **Q-Learning Algorithm**

Students need to implement **model-free Q-learning**:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [R_{t+1} + \rho \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)]$$

As there is no state variable, students only need to implement **stateless model-free Q-learning**:

$$Q(a_t) \leftarrow Q(a_t) + \alpha [R_{t+1} + \rho \max_{a'} Q(a') - Q(a_t)]$$

- **ϵ -greedy exploration** required

$$\epsilon_t = e^{-\beta t}$$

- Learning rate α , exploration decay β , and discount factor ρ :
 - Must be **carefully chosen**
 - Must be **reported and justified**

Convergence Requirement

- Simulate until the Q-values and/or average portfolio weights stabilize.
- Students must **clearly define**:
 - What constitutes convergence
 - How many iterations are required

Required Comparison: Q-Learning vs True Optimal Portfolio

After convergence, simulate T periods in steady state.

◦ Portfolio Weights

Compute average learned weights:

$$\bar{w}^{QL} = \frac{1}{T} \sum_t w_t^{QL}$$

Compare to w^* :

- bar plots
- Euclidean distance $\| \bar{w}^{QL} - w^* \|$, where w^* is the true optimal portfolio
- ...
- **Convergence Speed**
 - Plot cumulative reward difference relative to w^* , the true optimal portfolio
 - Report number of observations needed to approach optimality
 - ...

Evaluation Criteria

Controlled financial environment setup	$\approx 20\%$
Theoretical benchmark derivation	$\approx 20\%$
Q-learning algorithm description in detail	$\approx 20\%$
Q-learning implementation	$\approx 20\%$
Convergence analysis & comparison	$\approx 20\%$

Portfolio constraints such as no shorting => Significant Bonus Points!